

Interface Capability Brief

FTP Named Download Trigger · Stakeholder overview · 2026-08-12

1. Executive summary

Demonstrate how to download a remote file by a computed name when the FTP Operation processor has no Download operation (Move / Delete / Upload / List Files only). Uses stock PilotFish: parse a local control file → Listener Trigger (Run One Cycle) → separate FTP / SFTP Listener route that fetches the named file.

2. Who this is for

- Sources: Operators / Web UI dropping a .ctl control file into input/control - Remote store: demo SFTP (atmoz/sftp) with seeded payload + decoy - Destinations: downloaded bytes under output/downloaded/; control archive under output/archive/ - Demo vs production: Demo only (credentials demo/demo)

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3. What the interface can do

Capabilities below are drawn from the working design and, when present, the published CapabilityStatement — not from aspirational roadmap items.

- Transport: DirectoryListener polling \$\$CONTROL_INBOUND_DIRECTORY
- Format: plain-text control body = remote basename+extension (one line), e.g. invoice-20260407.dat
- Identity: control com.pilotfish.FileName; remote name attribute demo.ftp.RemoteFileName
- Samples: samples/control/fetch-invoice.ctl, samples/remote/invoice-20260407.dat

Integration highlights

Docs source: PilotFish Documentation 26R1.11 FTP Operation + Listener Trigger + FTP/SFTP Listener deep-dives.
Runtime image: pilotfish-eip:23R1 (call out version skew if config tags differ).

Stage	Module / FQCN	Notes
Route 1 listen	com.pilotfish.eip.modules.file.DirectoryListener	.ctl only; Move → archive
Save body → attr	com.pilotfish.eip.modules.internal.SaveDataToAttributeProcessor	Data Attribute Swapper → demo.ftp.RemoteFileNameRaw
Trim + path	com.pilotfish.eip.modules.internal.TransactionAttributePopulationProcessor	Sets demo.ftp.RemoteFileName, demo.ftp.RemoteFullPath
Trigger download	com.pilotfish.eip.modules.internal.ListenerTriggeringProcessor	RUN_ONCE=true → Route 2 FTP listener
Null transport	com.pilotfish.eip.modules.internal.NullTransport	Route 1 ends after trigger
Route 2 listen	com.pilotfish.eip.modules.file.ftp.FTPListener	JSCH SFTP; IS_TRIGGERABLE_LISTENER=true; UseFullPath + OGNL path

Stage	Module / FQCN	Notes
Write local	com.pilotfish.eip.modules.file.DirectoryTransport	\$\$DOWNLOAD_OUTPUT_DIRECTORY

4. Honest boundaries (not claimed yet)

- Optional synchronous wait / Call Route hand-back when a future build adds FTP Operation Download
- Optional FileNameRestriction directory-poll variant instead of UseFullFilePath

5. What you can verify in a walkthrough

Use these as checklist items when reviewing the demo with business stakeholders. They describe outcomes, not module names.

- Exercise route “1 - Parse Control And Trigger Download” from the Web UI or documented smoke path
- Exercise route “2 - Download Named File From SFTP” from the Web UI or documented smoke path
- Confirm outputs land in the documented folders / queues and appear in the Web UI status views

6. How it works (plain language)

7. Validation

- Control extension .ctl only - Exact remote path via FullPathToFile (not FTP Operation Download) - Decoy file on SFTP proves the listener does not grab everything

6. State & idempotency

- No SQL; success = file presence - FTP post-process = Keep so the seed can be re-fetched - Control Move to archive before download completes (accepted demo risk — trigger is async)

8. Dual-write / side effects

- Order: archive control (listener Move) → async FTP poll → write downloaded file - Compensation: none (demo)

3. Inbound contract

- Transport: DirectoryListener polling \$\$CONTROL_INBOUND_DIRECTORY - Format: plain-text control body = remote basename+extension (one line), e.g. invoice-20260407.dat - Identity: control com.pilotfish.FileName; remote name attribute demo.ftp.RemoteFileName - Samples: samples/control/fetch-invoice.ctl, samples/remote/invoice-20260407.dat

Routes in this interface

Each route is a configured PilotFish flow. Technical wiring appears in the Route Diagrams PDF; here is the stakeholder reading.

1 - Parse Control And Trigger Download: runs steps such as Poll Control Files, Save Control Body To Attribute, Set Remote File Name, Set Remote Full Path.

2 - Download Named File From SFTP: runs steps such as Download Named File From SFTP.FTP Listener, Write Downloaded File.

7. Security & trust posture

- Credentials and passwords in this package are demo-only — not production secrets
- Risk noted in design: Med — Listener Trigger Run One Cycle is async — control route finishes before download — Accepted; Web UI / tests wait on output file

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- Risk noted in design: Med — Archive-before-download — Accepted demo risk
 - Risk noted in design: Low — Docs 26R1.11 vs runtime 23R1 — Smoke-test modules; same FQCNs used in Sandbox/CRL
 - Risk noted in design: Low — Old JSCH vs modern OpenSSH — Custom sftp/sshd_config (same as EDI 835 OCI demo)
 - Risk noted in design: Info — FTP Operation still has no Download — Documented product gap; this demo is the workaround

8. Seeing it work

- - Ports: SFTP host 2223, EIP 8112, Web UI 8113 - LAN: LAN_HINT=http://192.168.68.62:8113/ - Compose project: ftp-named-download-trigger - Cold start ~60–90s for EIP
- docker compose up -d --build
- - Web UI local: http://localhost:8113/
- - Local PDF: http://localhost:8113/documents/route-diagrams.pdf

9. Related technical documents

- DESIGN.md — working engineering specification for this interface
- README.md — how to run, ports, and smoke commands
- This Capability Brief — shareable stakeholder summary (regenerate after design changes)

This document is generated from the interface sources listed above. If design and runtime diverge, believe the smoke-tested runtime and update DESIGN.md.